

Highlights

Label-free covariance alignment for longitudinal EEG decoding

- A 24,181-parameter EEG decoder emitting class and calibrated confidence.
- Label-free in-network alignment reduces session drift by 52–84% in 15 of 15 subjects.
- Test-time adaptation can track the streaming class and erase the signal it protects.
- Two cohorts have opposite adaptation-rate optima, costing 0.06–0.09 to get wrong.
- The admissible rate band is computable from the recording protocol before fitting.